

The ground, anxious not to disturb our homeward trajectory, told us not to dump any waste material overboard. What to do with urine taxed our ingenuity . . . I'm glad we got home when we did, because we were just about out of ideas for stowage.

A most remarkable achievement of Mission Control was quickly developing procedures for powering up the CM after its long cold sleep. They wrote the documents for this innovation in three days, instead of the usual three months. We found the CM a cold, clammy tin can when we started to power up. The walls, ceiling, floor, wire harnesses, and panels were all covered with droplets of water. We suspected conditions were the same behind the panels. The chances of short circuits caused us apprehension, to say the least. But thanks to the safeguards built into the command module after the disastrous fire in January 1967, no arcing took place. The droplets furnished one sensation as we decelerated in the atmosphere: it rained inside the CM.

Four hours before landing, we shed the service module; Mission Control had insisted on retaining it until then because everyone feared what the cold of space might do to the unsheltered CM heat shield. I'm glad we weren't able to see the SM earlier. With one whole panel missing, and wreckage hanging out, it was a sorry mess as it drifted away.

Three hours later we parted with faithful Aquarius, rather rudely, because we blasted it loose with pressure in the tunnel in order to make sure it completely cleared. Then we splashed down gently in the Pacific Ocean near Samoa, a beautiful landing in a blue-ink ocean on a lovely, lovely planet.

The Nixon administration was not interested in an ambitious post-Apollo program, and in early 1970 told NASA to plan for a much lower budget in coming years. This led to a series of NASA decisions on how best to adjust to a constrained budget outlook. George Low, who had become NASA deputy administrator in December 1969, announced on January 4, 1970, that NASA was canceling Apollo 20 and stretching out the remaining seven missions so that they would continue through 1974. Ten days later, faced with continuing budget cuts, administrator Thomas Paine announced that production of the Saturn V would be suspended indefinitely once the fifteenth vehicle had been

completed. Within six months of the first landing on the Moon, the United States had essentially abandoned the heavy-lift capability that was the key to future space exploration.

More cancellations were to come. NASA leadership was faced with how to fit both the cost of future Apollo missions and getting started on its post-Apollo programs into the reduced budget proposed by the White House. In addition, some within NASA, especially after the near-tragedy of Apollo 13, judged that the benefits of additional Apollo missions were not worth the risks involved. Reluctantly, NASA leadership agreed that NASA should cancel Apollo 15, the last mission without a lunar roving vehicle, and Apollo 19. The remaining flights after Apollo 14 were then renumbered Apollo 15 through 17. The White House was formally notified of the NASA decision in a September 1, 1970, letter to President Nixon.

**Letter to the President from T. O. Paine (Administrator, NASA),
September 1, 1970**

Dear Mr. President:

NASA has now completed an intensive review of the future of its manned flight program, including the six remaining Apollo lunar landing missions. We have determined that we should continue with four of the six previously planned lunar landings, but should drop the remaining two. It has been necessary to reach this decision at this time in order to avoid expenditures preparing for the Apollo 15 and 19 missions to be deleted, and to apply the necessary resources to those missions which are to be retained. During the conduct of this study, we requested and received the advice of the scientific community, and have kept your staff and the appropriate Congressional committees abreast of our effort.

The most compelling reason for the decision to delete these two flights, which we have arrived at reluctantly but with overwhelming consensus, is the current and reasonably foreseeable austere funding situation for NASA. As you know, we do not yet have our FY 1971 appropriation bill, but must proceed realistically to operate at a level not to exceed the \$3.268.7 million voted by Congress in the vetoed HUD and Independent Offices Bill. This is \$64 million below your budget request. In addition, a responsible and realistic view of the future indicates that increases

in the NASA budget over the next few years will probably be less than required to carry out all of your first priority goals in aeronautics and space. We estimate that the deletion of these two lunar landing missions will reduce NASA's expenditures by \$700-900 million over the next four years.

The scientific community, through our Lunar and Planetary Missions Board and the National Academy of Sciences' Space Science Board, strongly endorsed the retention of all six flights based on the scientific benefits to be derived. I fully understand and am sympathetic with their views, but these benefits do not, in our judgment, outweigh the benefits of other ongoing and future NASA programs and the risks involved in these difficult missions.

The six previously planned lunar flights would utilize all of the nation's available Saturn V launch vehicles, on which production was discontinued last year because of funding constraints. In view of Soviet progress on large launch vehicles, it is prudent to retain a modest Saturn V capability for such possible missions as launching a space station, retaining a back-up vehicle for scheduled missions, or for other unforeseen national requirements; deleting the Apollo 15 and 19 missions provides a national reserve of two Saturn V's.

You realize, I know, that each lunar mission is in itself a unique risk. We acknowledge this and must accept the responsibility. Obviously, there is an overall risk reduction in the dropping of two flights, and this advantage is multiplied by rescheduling the four remaining Apollo lunar landing missions to fly within a two-year period, without a significant gap between them. The momentum and morale engendered by such a compact schedule are distinct safety advantages in my view.

The proposed new manned flight program will fly the four remaining lunar flights at six-month intervals in 1971-72, followed by the Skylab program in which three teams of three astronauts will visit this space station prototype during 1973. There will then be a hiatus in the U.S. manned flight until the first test flights of the new space shuttle in 1975; these flights will continue through 1976, and the first manned orbital flight of the shuttle should be made in 1977. This program does represent a reduction in near-term manned flight, but it permits a sound development program for the space vehicle of the late 1970's. In light of the

existing and foreseeable near-term funding situation, it is clearly the most realistic and forward-looking approach to the nation's future in space and aeronautics.

As early as December 1969, Richard Nixon had questioned the need for eight additional Apollo missions, and by early 1971 was pressuring his associates to cancel the last two Apollo missions, Apollo 16 and Apollo 17. The near-tragic Apollo 13 reinforced Nixon's skepticism regarding the wisdom of continued flights to the Moon. Thomas Paine resigned as NASA administrator in September 1970. George Low served as acting administrator until his successor could be chosen. James Fletcher, a former president of the University of Utah and an industrial executive, became NASA administrator in May 1971. In November of that year, he wrote the deputy director of the Office of Management and Budget, Caspar W. Weinberger, to provide both reasons why canceling the final two Apollo missions was not a good idea and a rationale and needed actions if the president persisted in his attempts to cancel the missions.

James C. Fletcher (Administrator, NASA), Letter to Caspar W. Weinberger (Deputy Director, Office of Management and Budget), November 3, 1971

Dear Cap:

In our conversation last week, you indicated that cancellation of Apollo 16 and 17 was being considered by the President and asked for my views on the actions that should be taken to offset or minimize the adverse consequences if such a decision is made.

From a scientific standpoint these final two missions are extremely important, especially Apollo 17 which will be the only flight carrying some of the most advanced experiments originally planned for Apollos 18 and 19, cancelled last year. With what we have learned from Apollo 15 and previous missions, we seem to be on the verge of discovering what the entire moon is like: its structure, its composition, its resources, and perhaps even its origin. If Apollo 16 and 17 lead to these discoveries, the Apollo program will go down in history not only as man's greatest adventure, but also as his greatest scientific achievement. Recognizing the great scientific potential and the relatively small saving (\$133 million)

compared to the investment already made in Apollo (\$24 billion), I must as Administrator of NASA strongly recommend that the program be carried to completion as now planned.

If broader considerations, nevertheless, lead to a decision to cancel Apollo 16 and 17, the consequences would be much more serious than the loss of a major scientific opportunity. Unless compensatory actions are taken at the same time to offset and minimize the impact, this decision could be a blow from which the space program might not easily recover. As you requested, I will summarize the principal adverse consequences as I see them and then outline my recommendations on the compensatory actions necessary.

PRINCIPAL ADVERSE CONSEQUENCES

1. Negative Effect on Congressional and Public Support.

Without strong compensatory actions, a decision to cancel Apollo 16 and 17 would undermine the support the space program now enjoys and jeopardize the continued support that is required over the years to sustain the nation's position in space. Even though enthusiasm for the space program has diminished since the first lunar landing, NASA has continued to receive better than 98 percent of its budget requests each year (99.94% in FY 1972) because a substantial majority has accepted the judgments of the Administration and NASA's leadership that the space program is vital to the United States and that the programs recommended each year are necessary to achieve our goals. Cancellation of Apollo 16 and 17 would undermine this support in two ways.

First, it would call into question our credibility on this and other major elements of the space program since it would be a sudden reversal of the position we have so recently strongly supported in defense of our FY 1972 budget.

Second, it would terminate our best known, most visible and most exciting program which, in the minds of many in Congress and the public, has been the symbol of the space program and its success.

These factors, unless offset by strong positive actions, could result in a loss of confidence and interest that would have a

"domino" effect, causing us to lose support for the programs which are essential to the long-term future of the nation in space.

2. Impact on Science and the Scientific Community.

At this time, the entire cognizant scientific community is strongly in favor of Apollo 16 and 17. Cancellation would come as a shock and a surprise in view of the strong support these missions have received from the President's Science Adviser, all of NASA's science advisory groups, NASA management, and the Congress. There will be strong and vocal critical reaction.

3. Impact on Industry.

Taken by itself, the direct impact of the cancellation of Apollo 16 and 17 would be further reductions in 1972 of over 6,000 aerospace jobs. The hardest hit areas would be Southern California, Long Island, Cape Kennedy, and Houston. Unless the decision is coupled with commitments and actions to proceed with and possibly expedite other programs, like the space shuttle, it will be a devastating blow, actually and psychologically, to an already hard-hit industry.

4. Impact on NASA.

The impact on NASA will be felt most strongly at Houston and, to a lesser extent, at Huntsville and Cape Kennedy. A major problem will be to hold together for over a year the team we will need to rely on to conduct safely the Skylab missions in 1973. We will have to deal with the difficult and visible problem of the futures of the 16 astronauts now assigned to Apollo 16 and 17. The blow to morale throughout NASA will be serious unless, again, the decision is coupled with clear decisions and commitments on future programs.

5. Impact on the Public.

The large segment of space enthusiasts in the population at large would be extremely disappointed by the proposed cancellation.

Included in this group would be millions who have come to Cape Kennedy, often from very long distances, to witness Apollo launches, and the much larger numbers who follow each mission closely on TV. These groups may be a minority in the U.S. but they are quite vocal and certainly non-negligible in size.

6. Impact Abroad.

It is our understanding from USIA reports that the Apollo flights have been a major plus factor for the U.S. image abroad. The impact of cancelling Apollo 16 and 17 should be assessed in arriving at a decision.

RATIONALE AND ACTIONS REQUIRED

If a decision is made to cancel Apollo 16 and 17, it is essential to provide a clearly stated and defensible rationale and take constructive actions to minimize the adverse impacts of the cancellation on the space program, the Administration, and the individuals, "communities of interest," and organizations affected. The rationale and actions must make it clear that, in spite of the cancellation, the President continues to support a program involving man in space and with strong scientific content.

Rationale

The rationale supporting this position would be as follows:

"Our space program has three basic purposes: exploration; the acquisition of scientific knowledge; and practical applications for man on earth. (See President's statement of March 7, 1970.) We must always strive to achieve the proper balance among these purposes.

"Today we must stress two aspects of our space program. We must give a top priority to practical applications now possible and press forward with the development of earth oriented systems which will enable us to make wider and more effective practical uses of space in the future.

"The key to the future in space—in science and exploration as

well as practical applications—is routine access to space. Space activities will be part of our lives for the rest of time. These activities cannot continue, for long, to be as complex, as demanding, or as costly as they are today. We must develop new, simpler, less expensive techniques to go to space and to return from space. This is the goal of the space shuttle program. The sooner we get on with this development, the sooner will we be able to turn our knowledge gained in space science and space exploration toward helping man on earth.

"To operate in space most effectively we must also learn more about how man can best live and work in space. So while we are developing the shuttle, we must conduct space operations over longer periods of time—with Skylab.

"But to do all these things within limited resources, we must give up something. And when all factors are considered, the best project to give up—most reluctantly—is the remainder of Apollo: Apollo 16 and 17. This will for a time curtail our program of manned exploration and science.

"But we will, of course, continue exploration deep into space with unmanned spacecraft, including a landing on Mars in July 1976 with Viking, and the exploration of all the outer planets, Jupiter and beyond, with the Grand Tour late in this decade. The unmanned science program, with its High Energy Astronomy Observatory and other spacecraft will also continue to expand our fundamental knowledge of the universe. It is only manned science, and manned exploration, that will be curtailed.

"The United States must continue to fly men in space. Man will fly in space, and we cannot forego our responsibility—to ourselves and to the free world—to take part in this great venture. But for a time man can devote his own efforts, from space, toward practical needs here on earth, while leaving exploration beyond the earth to machines."

CONCLUSIONS

I recommend against the cancellation of Apollo 16 and 17 because these flights are scientifically important, and because much of the overall support for NASA's space program depends on our actions with respect to these flights.

If, nevertheless, for reasons external to NASA, Apollo 16 and 17 must be cancelled, then it becomes necessary to:

1. Provide strong backing to the manned earth-oriented space program.
2. Develop a rationale for the actions taken that is credible and supportable.
3. Take compensatory actions that will minimize the impact on the remaining NASA programs and their support.

The proposed rationale for the cancellation of Apollo 16 and 17 is that, in these times of pressing domestic needs, the manned space program should be earth-oriented instead of exploration and science-oriented.

The compensatory actions involve an early go-ahead for the space shuttle, the inclusion of "gap-filler" missions between Skylab and the shuttle, a number of augmented unmanned space science programs, and maintaining a total NASA budget at the FY 1971-1972 level of about \$3.3 in budget authority.

Nixon's feelings on the two remaining Apollo missions were captured in a recording of a November 24, 1971, meeting discussing various budget issues among the president, his domestic policy adviser John Ehrlichman, his budget director George Shultz, and Secretary of the Treasury John Connally. Nixon did not want an Apollo launch to take place in the run-up to the November 1972 election. He was persuaded after this meeting to allow the Apollo 16 launch, which was moved up from its original launch date in mid-1972 to April 1972 so that it would not take place in close proximity to the election.

Recording of Oval Office Conversation, November 24, 1971

Nixon: Space, what's the problem here?

Ehrlichman: Well the problem here is do we go ahead with the next two shots [Apollo 16 and 17]?

Nixon: No! If we go, no shots before the election.

Ehrlichman: Then what would we do with all those employees?

Nixon: For those shots? How many, George?

Shultz: 17,500 or something like that.

Nixon: I don't like the feeling of space shots between now and the election.

Ehrlichman: But thinking of this thing [the space program] in just pure job terms, it is a hell of a job creator.

Connally: The American people are really not impressed by any more space shots.

Nixon: NASA is saying you'll find incredible things about the moon with these last two shots, and the American people say "so what?"

Shultz: Could I try another possibility? The last shot is the one in which they have loaded a great amount of scientific stuff from the ones that have been canceled before. That shot is scheduled after the election.

Nixon: I only see a minor waste of money. Keep the people on, but don't make the shots. I just don't feel the shots are a big deal at this time . . . There is also the risk you could have another Apollo 13 . . . That would be the worst thing we could have . . . We are just not going to do it. There will not be any launches between now and the election. The last shot, fine. Let's go forward with the last shot.

On August 13, 1971, NASA had announced that the crew for the last mission to the moon, Apollo 17, would include as lunar module pilot Harrison H. "Jack" Schmitt, a PhD geologist who had come to NASA as a scientist-astronaut in 1965 and had been deeply involved in planning the science to be done on the lunar missions. This assignment was the result of pressure from the scientific community, who argued that the Apollo missions should include at least one trained scientist as a crewmember.

Schmitt's selection came on the heels of the scientifically most successful mission to date, Apollo 15, launched on July 26, 1971. This was the first mission to carry the lunar roving vehicle, and astronauts David Scott and James Irwin used the vehicle to traverse almost seventeen miles of the lunar surface, a distance much greater than that traveled by the previous three crews. The crew spent three days on the Moon and conducted three exploration sorties using their lunar roving vehicle activities. Most significant, they identified and brought back to Earth specimens of the primitive lunar crust, the first material that had solidified

from the molten outer layer of the young moon; one of these samples was dubbed the "Genesis rock."

The penultimate Apollo mission, Apollo 16, was launched on April 16, 1972. The mission landed in the lunar highlands, an area of the Moon that had not yet been explored. Apollo 16's objectives were similar to those of the preceding mission, with a focus on characterizing a region thought to be representative of much of the lunar surface.

All of the prior Apollo missions had been launched during daylight hours. After an almost three-hour delay, Apollo 17 lifted off at 12:33 a.m. EST on December 7, 1972. The vivid light from the Saturn V's five F-1 engines illuminated the night sky with an unreal brilliance. Astronauts Gene Cernan and Jack Schmitt spent three days exploring the lunar surface. As they prepared to leave the Moon, Cernan unveiled a plaque on the descent stage of the lunar module, which would remain on the Moon's surface. It read: "Here man completed his first explorations of the moon." As he took a last look at the lunar landscape, Cernan added, "As we leave the moon at Taurus-Littrow, we leave as we came, and, God willing, as we shall return, with peace and hope for mankind." The lunar module Challenger lifted off from the Moon at 5:55 p.m. EST on December 14.

With its departure, a remarkable era in human history came to a close. For the first time, human beings had left their home planet. In a statement issued after Apollo 17 had left the Moon, Richard Nixon declared, "This may be the last time in this century that men will walk on the Moon." With the decisions he made in the 1969-72 period, Nixon turned this prophecy into reality.

Richard Nixon, "Statement After Lift-Off from the Moon of Apollo 17 Lunar Module," December 14, 1972

As the Challenger leaves the surface of the Moon, we are conscious not of what we leave behind, but of what lies before us. The dreams that draw humanity forward seem always to be redeemed if we believe in them strongly enough and pursue them with diligence and courage. Once we stood mystified by the stars; today we reach out to them. We do this not only because it is man's destiny to dream the impossible, to dare the impossible, and to do the impossible, but also because in space, as on Earth,

there are new answers and new opportunities for the improvement and the enlargement of human existence.

This may be the last time in this century that men will walk on the Moon. But space exploration will continue, the benefits of space exploration will continue, the search for knowledge through the exploration of space will continue, and there will be new dreams to pursue based on what we have learned. So let us neither mistake the significance nor miss the majesty of what we have witnessed. Few events have ever marked so clearly the passage of history from one epoch to another. If we understand this about the last flight of Apollo, then truly we shall have touched a "many-splendored thing."

To Gene Cernan, Jack Schmitt, and Ronald Evans, we say God speed you safely back to this good Earth.